# One ideal-class element: fixed bound 200

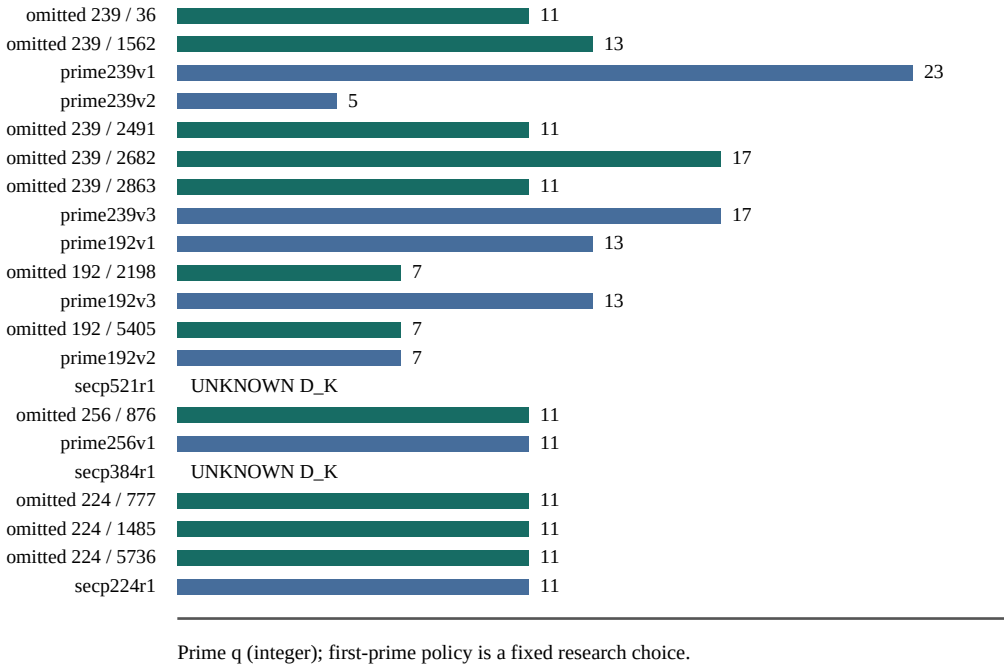
28 October 2026 UTC | T-5518 | NIST-CLASS-ELEMENT-ORDER-01 | observed exact arithmetic

**All eleven omissions pass on the first witness.** The first odd split prime gives an ideal class of order at least 200 for eleven omissions and eight publications. Both methods agree on 3,781 reduced powers per arm. Published P-384/P-521 remain UNKNOWN_D_K; all 21 rows and six streams are retained.

| Class number bound | Individual class element | Historical use |
|---|---|---|
| 19 fields: h(K) >= 2^24 | 19 fields: order >= 200 | 1997 caller: unknown |

## First odd split prime used for the single-element witness

Teal: omitted; blue: published. Every known-D_K row passes order >= 200.




Prime q (integer); first-prime policy is a fixed research choice.

**What changed.** ETSI (July 2005), Annex D.1/D.3, asks for one element of order at least 200, and leaves element selection and retry limits unspecified. A large class number alone does not prove a large-order individual element. This test checks the latter directly, using a fixed first-prime policy before observing any cohort powers.

**Lifting limit.** The cited rationale concerns a characteristic-zero lift preserving the CM endomorphism ring. Lifting equation coefficients to integers does not establish that property or a discrete-log transfer. No 1997 NIST adoption or historical rejection policy follows.

Primary source: ETSI TS102176-1 V1.2.1 (2005-07), Annex D.1/D.3, printed 43-46. d-nb.info/964515040/34 section2.3.5 gives the 2002 discussion; PARI Arithmetic_functions.html documents exact form composition. Full URLs, paths and limits are in report.md.

# Full cohort, independent arithmetic and limits

| Curve / block | Label | q | Powers | Element order |
|---|---|---|---|---|
| omitted 239 / 36 | omitted | 11 | 199 | >=200 |
| omitted 239 / 1562 | omitted | 13 | 199 | >=200 |
| prime239v1 | published | 23 | 199 | >=200 |
| prime239v2 | published | 5 | 199 | >=200 |
| omitted 239 / 2491 | omitted | 11 | 199 | >=200 |
| omitted 239 / 2682 | omitted | 17 | 199 | >=200 |
| omitted 239 / 2863 | omitted | 11 | 199 | >=200 |
| prime239v3 | published | 17 | 199 | >=200 |
| prime192v1 | published | 13 | 199 | >=200 |
| omitted 192 / 2198 | omitted | 7 | 199 | >=200 |
| prime192v3 | published | 13 | 199 | >=200 |
| omitted 192 / 5405 | omitted | 7 | 199 | >=200 |
| prime192v2 | published | 7 | 199 | >=200 |
| secp521r1 | published | ? | 0 | UNKNOWN D_K |
| omitted 256 / 876 | omitted | 11 | 199 | >=200 |
| prime256v1 | published | 11 | 199 | >=200 |
| secp384r1 | published | ? | 0 | UNKNOWN D_K |
| omitted 224 / 777 | omitted | 11 | 199 | >=200 |
| omitted 224 / 1485 | omitted | 11 | 199 | >=200 |
| omitted 224 / 5736 | omitted | 11 | 199 | >=200 |
| secp224r1 | published | 11 | 199 | >=200 |

**Independent paths.** Python constructs prime-power ideals by Hensel lifting and elementary Gauss reduction. PARI repeatedly composes the original form. Every reduced triple agrees, through powers 1..199; zero identity returns on the 19 nonmissing rows. Input discriminants are rechecked; GP proves source-factor primality. Complete models, UIDs and transcripts are linked in cohort.csv.

**Controls and process outcomes.** Exact orders 1, 2, 3, passing/failing threshold controls, nonfundamental input and deterministic A/A pass. Exception and early-return guards fail closed. Two control-only GP parser failures and their repair precede freeze and remain retained. Both whole-cohort arms complete; no unexpected trial failure, timeout or retry. Export file verification passes. Desktop clocks are diagnostic only.

**Boundary.** First-prime selection is a fixed research policy; ETSI does not prescribe it. These observations exclude no omissions at threshold 200, but establish no original NIST invocation, retry/error policy or historical class-group implementation. Neither class number nor this element-order property identifies the missing selector. The P-224 splitting lead and other source-defined mechanisms remain open.

Frozen protocol 0636225ef6ce97eaa0c831093b946a6da960edd0e04c366d12db65e68f86ec60; freeze commit a49aa5259d; raw commit c8163cfe2b. Exact commands, source/runtime hashes, controls and all 42 per-curve transcript files are retained. Historical goal active/unresolved; no official status promotion.